2026

GREENCAMPUS AI

AI POWERED CAMPUS-SUSTAINABILITY MONITORING SYSTEM

1M1B AI FOR SUSTAINABILITY VIRTUAL INTERNSHIP

IN COLLABORATION WITH IBM SKILLSBUILD & AICTE

SDG 12 – RESPONSIBLE CONSUMPTION AND PRODUCTION

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1. Abstract

GreenCampus AI is an AI-powered campus sustainability monitoring system designed to help educational institutions understand and manage their resource consumption.

College campuses use resources such as electricity and water every day and also generate waste. However, without systematic monitoring, it can be difficult to identify unusual consumption patterns and understand where resources may be getting wasted.

The proposed system uses simulated campus data containing electricity consumption, water consumption, waste generation, occupancy, temperature, working days and event days. Python is used for data preparation, analysis and AI models, while Power BI is used to create an interactive sustainability dashboard.

The system uses AI for **pattern detection, anomaly detection, electricity prediction and decision support**. Anomaly detection identifies unusual resource-consumption patterns, while the recommendation system suggests areas that should be investigated.

The project follows responsible AI principles by maintaining transparency, protecting privacy, considering contextual differences between buildings and keeping humans involved in final decisions.

2. Introduction

Sustainability has become an important concern for educational institutions. A college campus can contain classrooms, laboratories, libraries, hostels and other facilities that consume different amounts of electricity and water.

Resource consumption is affected by several factors such as:

- Number of people using a building
- Building type
- Temperature
- Working days
- Events
- Daily activities

If this information is collected and analyzed properly, it can help the campus understand its resource-use patterns.

GreenCampus AI was developed as a decision-support system that uses data analysis and AI to monitor these patterns and highlight unusual situations.

The project follows the internship requirement of starting with a real-world sustainability problem and applying AI where it can improve analysis, monitoring and decision-making. The guideline also identifies prediction, pattern detection and decision support as suitable AI roles.

3. Sustainable Development Goal

SDG 12 – Responsible Consumption and Production

GreenCampus AI is mainly aligned with **SDG 12**, which focuses on responsible consumption and production.

The project supports this goal by helping a campus:

- Monitor electricity consumption
- Monitor water consumption
- Monitor waste generation
- Identify unusual resource-use patterns
- Understand consumption trends
- Support data-driven sustainability decisions
- Increase awareness about responsible resource usage

The project focuses on **better monitoring and decision support** rather than claiming that the system itself directly reduces resource consumption.

4. Problem Statement

Official Problem Statement

How might we use AI to monitor and analyze campus resource consumption so that college campuses can reduce resource wastage and become more sustainable?

Problem Description

College campuses use large amounts of resources every day. Electricity is required for lighting, computers, laboratories and cooling systems. Water is required for drinking, cleaning, sanitation and other activities. The campus also generates waste from daily activities and special events.

Without proper monitoring, unusual consumption may not be noticed quickly.

For example:

- Electricity consumption may suddenly become unusually high.
- Water usage may increase because of a possible leakage or unusual activity.
- Waste generation may increase during events.
- Different buildings may have different normal consumption levels.

Therefore, there is a need for a system that can analyze resource data and highlight unusual patterns for further investigation.

5. Objectives

The main objectives of GreenCampus AI are:

1. To monitor campus electricity consumption.
2. To monitor campus water consumption.
3. To monitor waste generation.
4. To analyze resource-consumption patterns.
5. To identify unusual consumption using AI.
6. To predict electricity consumption.
7. To generate AI-based investigation recommendations.
8. To present important information through a Power BI dashboard.
9. To support data-driven sustainability decisions.
10. To apply responsible AI principles.

6. Target Users

The proposed system can be useful for different campus stakeholders.

User	Possible Use
College Administration	Monitor overall resource consumption
Facility/Maintenance Staff	Investigate unusual resource usage
Faculty	Understand sustainability patterns
Students	Increase awareness about sustainable resource use

The main users are expected to be **college administration and facility/maintenance staff**, because they can use the information for monitoring and investigation.

7. Proposed Solution

GreenCampus AI follows a data-driven AI workflow.

The overall process is:

Campus Resource Data
↓
Data Cleaning
↓
Exploratory Data Analysis
↓
AI Anomaly Detection
↓
Electricity Prediction
↓
AI Recommendation Generation
↓
Power BI Dashboard
↓
Human Review and Investigation

The system does not automatically make decisions. Instead, it provides information and recommendations that can be reviewed by humans.

8. Dataset

For this project, a **simulated/synthetic dataset** was created because actual college resource-consumption data was not used.

The dataset represents one year of campus activity.

Dataset Details

Parameter	Details
Data type	Simulated/Synthetic
Period	July 2025 – June 2026
Number of buildings	5
Number of records	1,825
Resources	Electricity, Water, Waste

Buildings Used

1. Block A
2. Block B
3. Block C
4. Library
5. Hostel

Different building types and occupancy levels were used to make the simulated dataset more realistic.

9. Dataset Attributes

The dataset contains the following attributes:

Column	Description
Date	Date of observation
Building	Campus building
Building_Type	Type of building
Occupancy	Number of people represented for the day
Electricity_kWh	Electricity consumption
Water_Liters	Water consumption
Waste_kg	Waste generated
Temperature_C	Temperature
Working_Day	Whether it is a working day
Event_Day	Whether a campus event occurred

The dataset was designed so that resource consumption could vary according to factors such as occupancy, temperature, working days and events.

10. Data Generation

Python was used to generate the simulated dataset.

The dataset contains:

- 365 days
- 5 buildings
- 1,825 total observations

The generated data was designed to represent reasonable relationships between campus activities and resource consumption.

For example:

- Higher occupancy can result in higher resource consumption.
- Higher temperature can increase electricity requirements.
- Non-working days can have lower consumption.
- Events can increase occupancy and waste generation.

```
df.shape
✓ [9] 62ms
(1825, 10)
```

```
df.head(5)
✓ [10] 39ms
```

	Date	Building	Building_Type	Occupancy	Electricity_kWh	Water_Litres	Waste_kg	Temperature_C	Working_Day	Event_Day
0	2025-07-01	Block A	Academic	444	1437.09	8384.17	47.61	28.0	Yes	No
1	2025-07-01	Block B	Academic	269	1091.78	4583.04	23.77	28.0	Yes	No
2	2025-07-01	Block C	Laboratory	186	707.71	3556.96	22.42	28.0	Yes	No
3	2025-07-01	Library	Library	230	760.70	4039.62	24.38	28.0	Yes	No
4	2025-07-01	Hostel	Hostel	555	1679.32	9611.07	54.21	28.0	Yes	No

Figure 10.1 Simulated campus sustainability dataset generated using Python

11. Data Cleaning and Validation

After generating the dataset, data cleaning and validation were performed.

The following checks were carried out:

- Missing values
- Duplicate records
- Data types
- Negative values
- Unique building names
- Building types
- Working-day values
- Event-day values
- Date range

The dataset was also checked to ensure that the intentionally created unusual cases were retained.

The cleaned dataset was then saved for further analysis.

```
df.isnull().sum()
✓ [11] 38ms

Date              0
Building          0
Building_Type     0
Occupancy         0
Electricity_kWh   0
Water_Litres      0
Waste_kg          0
Temperature_C     0
Working_Day       0
Event_Day         0
dtype: int64
```

Figure 11.1 Missing value check

```
df.duplicated().sum()
✓ [12] 126ms
np.int64(0)
```

Figure 11.2 Duplicate check

```
df.dtypes
✓ [4] 249ms
Date              object
Building          object
Building_Type     object
Occupancy          int64
Electricity_kWh  float64
Water_Litres     float64
Waste_kg         float64
Temperature_C    float64
Working_Day       object
Event_Day         object
dtype: object
```

Figure 11.3 Data types

```
df.to_csv("cleaned_campus_sustainability_data.csv", index=False)

print("Cleaned dataset saved successfully!")
✓ [9] 110ms
Cleaned dataset saved successfully!
```

```
df.head()
✓ [8] 85ms
```

	Date	Building	Building_Type	Occupancy	Electricity_kWh	Water_Litres	Waste_kg	Temperature_C	Working_Day	Event_Day
0	2025-07-01	Block A	Academic	444	1437.09	8384.17	47.61	28.0	Yes	No
1	2025-07-01	Block B	Academic	269	1091.78	4583.04	23.77	28.0	Yes	No
2	2025-07-01	Block C	Laboratory	186	707.71	3556.96	22.42	28.0	Yes	No
3	2025-07-01	Library	Library	230	760.70	4039.62	24.38	28.0	Yes	No
4	2025-07-01	Hostel	Hostel	555	1679.32	9611.07	54.21	28.0	Yes	No

Figure 11.4 Final dataset shape

12. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the dataset before applying AI models.

The analysis included:

12.1 Building-wise Analysis

Resource consumption was compared across different campus buildings.

This helps identify buildings with higher or lower resource usage.

12.2 Monthly Analysis

Monthly trends were analyzed to understand how consumption changes over time.

12.3 Occupancy Analysis

The relationship between occupancy and resource consumption was studied.

12.4 Temperature Analysis

Temperature was compared with electricity consumption.

12.5 Working-Day Analysis

Resource consumption on working and non-working days was compared.

12.6 Event-Day Analysis

Event days were analyzed to understand changes in occupancy and waste generation.

```
monthly_electricity.plot(kind="line", marker="o")

plt.title("Monthly Campus Electricity Consumption")
plt.xlabel("Month")
plt.ylabel("Electricity (kWh)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
[13]
```

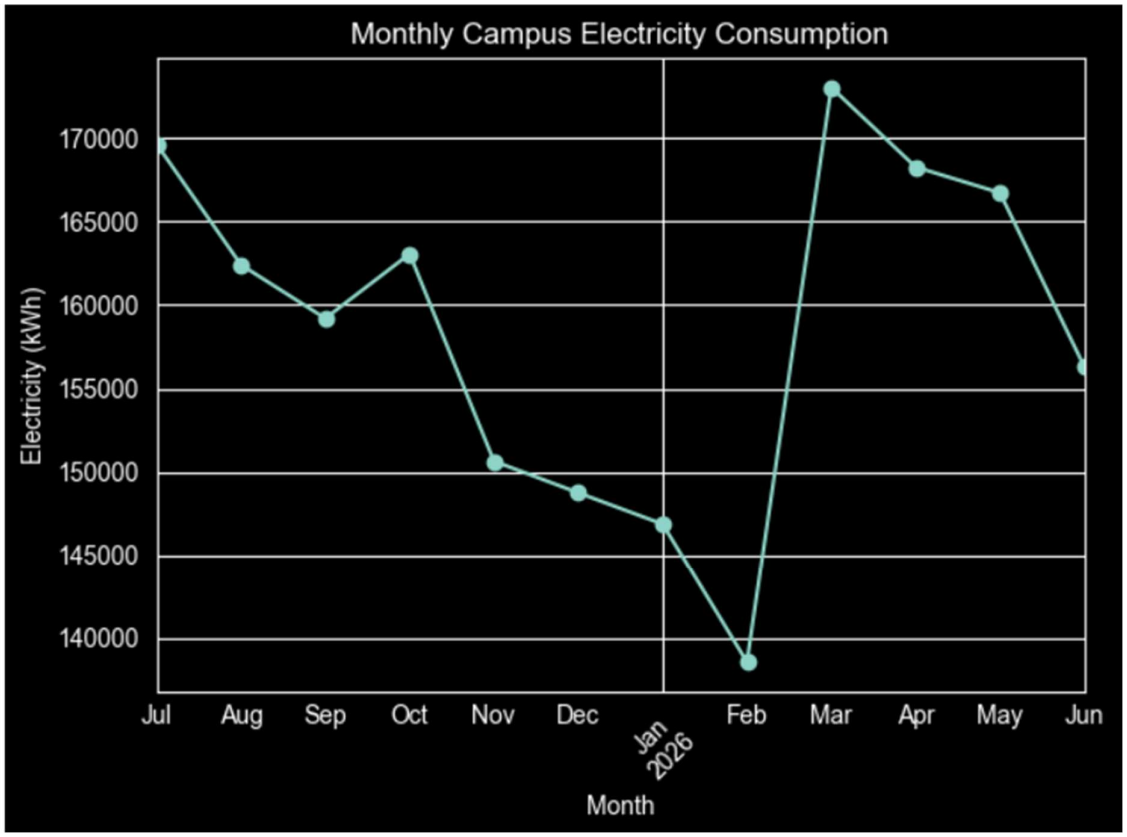


Figure 12.1 Exploratory Analysis of Campus Electricity Consumption

```
plt.scatter(df["Occupancy"], df["Electricity_kWh"])

plt.title("Occupancy vs Electricity Consumption")
plt.xlabel("Occupancy")
plt.ylabel("Electricity (kWh)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

[16]

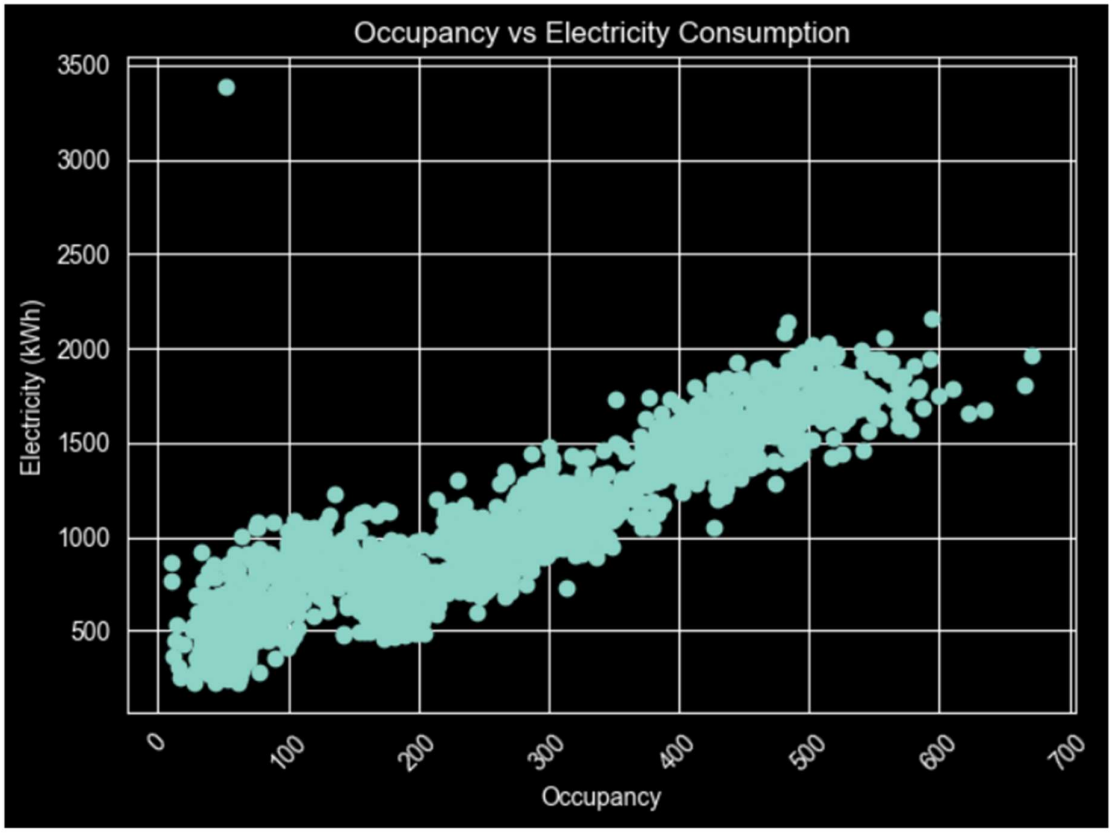


Figure 12.2 Relationship between Occupancy and Electricity Consumption

13. AI-Based Anomaly Detection

One of the main AI components of GreenCampus AI is anomaly detection.

The purpose of anomaly detection is to identify observations that appear significantly different from normal resource-consumption patterns.

Algorithm Used

Isolation Forest

Isolation Forest is an unsupervised machine-learning algorithm that can identify unusual observations in a dataset.

The following features were used:

- Occupancy
- Electricity
- Water
- Waste
- Temperature

The model assigns each record as either:

- **Normal**
- **Anomaly**

An anomaly score was also calculated to help identify records that were more unusual.

14. Intentional Anomalies

To test the anomaly-detection system, several unusual cases were intentionally introduced into the simulated dataset.

Examples include:

Date	Building	Resource
15 February 2026	Block A	Electricity
10 April 2026	Block B	Water
20 May 2026	Library	Waste

These cases were created to test whether the AI system could identify unusual resource-consumption patterns.

Importantly, an AI anomaly does **not automatically prove the reason for the anomaly**. It only indicates that the observed pattern is unusual.

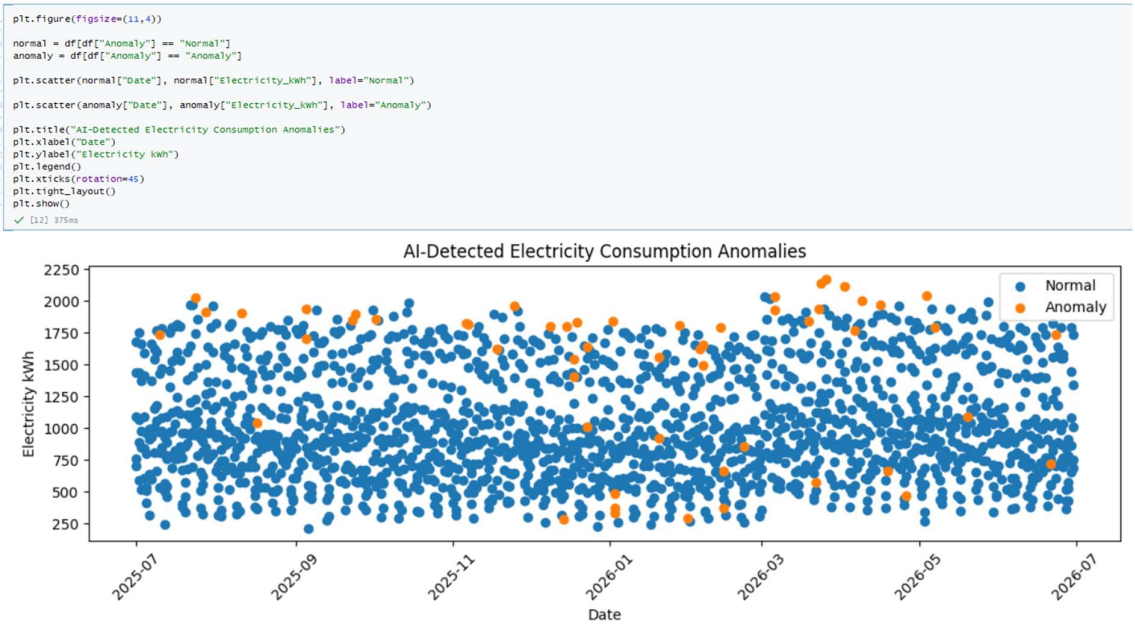


Figure 13.1 AI-based anomaly detection using Isolation Forest

15. AI-Based Electricity Prediction

A second AI component was developed to predict electricity consumption.

Algorithm Used

Random Forest Regression

The model uses different campus factors to estimate electricity consumption.

Features include:

- Occupancy
- Water consumption
- Waste generation
- Temperature
- Month
- Day of week
- Working-day status
- Event-day status
- Building
- Building type

The dataset was divided into training and testing data.

The Random Forest model was then trained using the training data and evaluated using the test data.

16. Prediction Evaluation

The prediction model can be evaluated using:

MAE – Mean Absolute Error

Shows the average difference between actual and predicted electricity consumption.

RMSE – Root Mean Squared Error

Measures prediction error while giving greater importance to larger errors.

R^2 Score

Shows how well the model explains variation in the target variable.

MAE: 66.6
RMSE: 86.5
R^2 Score: 0.961

```
plt.figure(figsize=(10,5))

plt.plot(y_test.values[:100], label="Actual")

plt.plot(y_pred[:100], label="Predicted")

plt.title("Actual vs Predicted Electricity Consumption")
plt.xlabel("Test Record")
plt.ylabel("Electricity Consumption (kWh)")
plt.legend()
plt.show()
✓ [38] 220ms
```

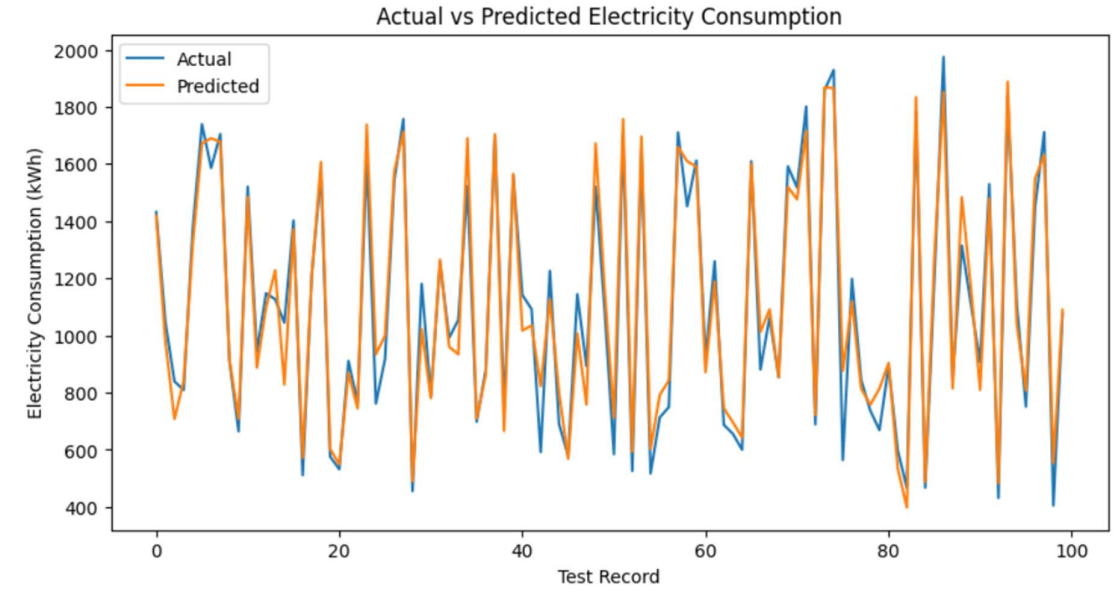


Figure 16.1 Electricity consumption prediction using Random Forest Regression

17. AI Decision Support and Recommendations

After anomaly detection, the system generates recommendations for further investigation.

The recommendation system uses resource-consumption thresholds and the anomaly result.

Examples include:

Electricity

“Unusually high electricity consumption detected. Investigate equipment usage, cooling demand, operating hours, or possible energy wastage.”

Water

“Unusually high, water consumption detected. Investigate possible leakage, meter issues, or unusual water usage.”

Waste

“Unusually high waste generation detected. Review event activity, waste segregation, and disposal practices.”

The recommendations are intentionally written as **investigation suggestions** rather than definite conclusions.

For example, the system does not say:

“There is definitely a water leak.”

Instead, it says that unusually high, water consumption should be investigated.

```
anomalies = df[df["Anomaly"] == "Anomaly"]

anomalies[["Date", "Building", "Electricity_kWh", "Water_Litres", "Waste_kg", "AI_Recommendation"]].head(20)
✓ [6] 59ms
```

	Date	Building	Electricity_kWh	Water_Litres	Waste_kg	AI_Recommendation
79	2025-07-16	Hostel	1854.50	10095.70	81.11	Unusually high electricity consumption detecte...
154	2025-07-31	Hostel	2017.45	8911.06	51.29	Unusually high electricity consumption detecte...
249	2025-08-19	Hostel	1801.08	10354.48	60.02	Unusually high electricity consumption detecte...
259	2025-08-21	Hostel	1922.63	9844.31	58.27	Unusually high electricity consumption detecte...
290	2025-08-28	Block A	1423.51	9314.49	63.68	Unusually high waste generation detected. Revi...
294	2025-08-28	Hostel	1847.84	9732.12	73.00	Unusually high electricity consumption detecte...
319	2025-09-02	Hostel	1959.72	12850.21	84.27	Unusually high electricity consumption detecte...
364	2025-09-11	Hostel	1748.32	10476.07	64.73	Unusually high water consumption detected. Inv...

Figure 17.1 AI-generated recommendations for unusual resource-consumption patterns

18. Power BI Dashboard

Power BI was used to convert the analysis results into an interactive dashboard.

The dashboard contains **two pages**.

19. Dashboard Page 1 – Sustainability Overview

The first dashboard page provides an overall view of campus sustainability.

It includes:

- Total Electricity
- Total Water
- Total Waste
- Average Occupancy
- AI Anomalies
- Date filter
- Electricity by Building
- Water by Building
- Electricity Trend
- AI Detected Resource Patterns
- AI Anomalies by Building

This page helps users quickly understand overall resource consumption and unusual patterns.

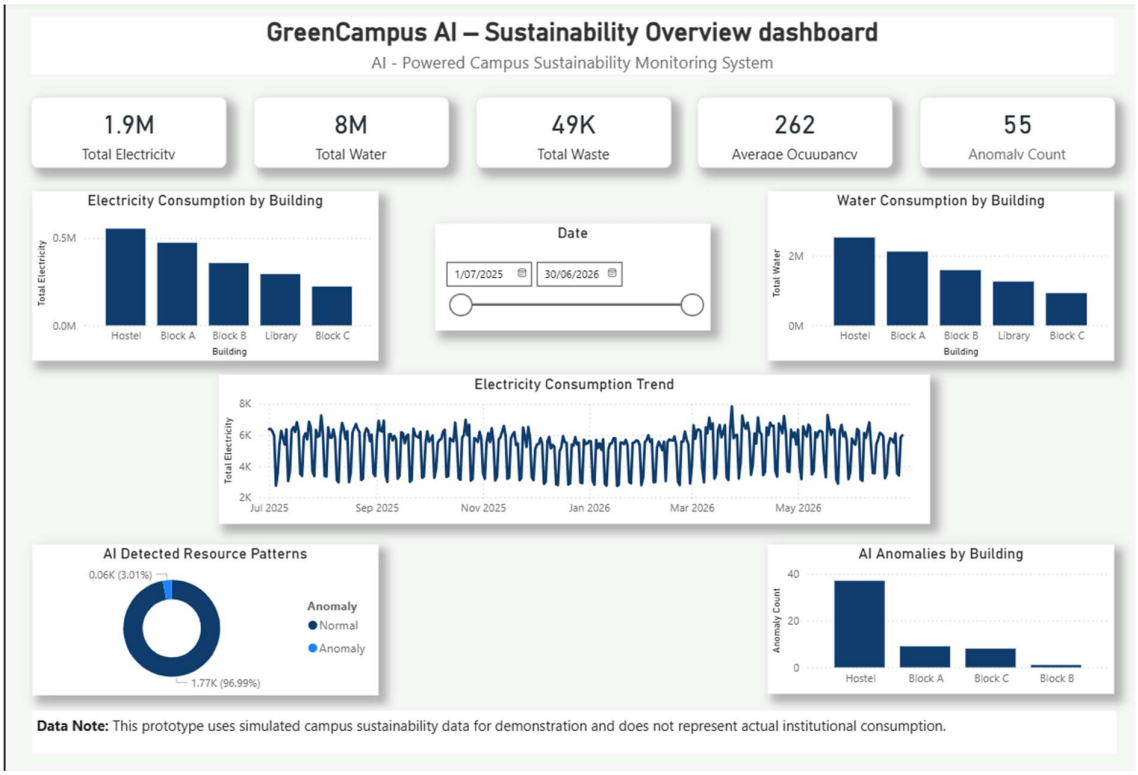


Figure 19.1 GreenCampus AI – Sustainability Overview dashboard

20. Dashboard Page 2 – AI Recommendations

The second dashboard page focuses on AI-based decision support.

It contains:

- AI Recommendation Categories
- AI Alerts by Building
- AI recommendation table
- Date
- Building
- Building type
- Electricity consumption
- Water consumption
- AI recommendation
- Anomaly status

The table allows users to see individual unusual observations and the corresponding recommendation.

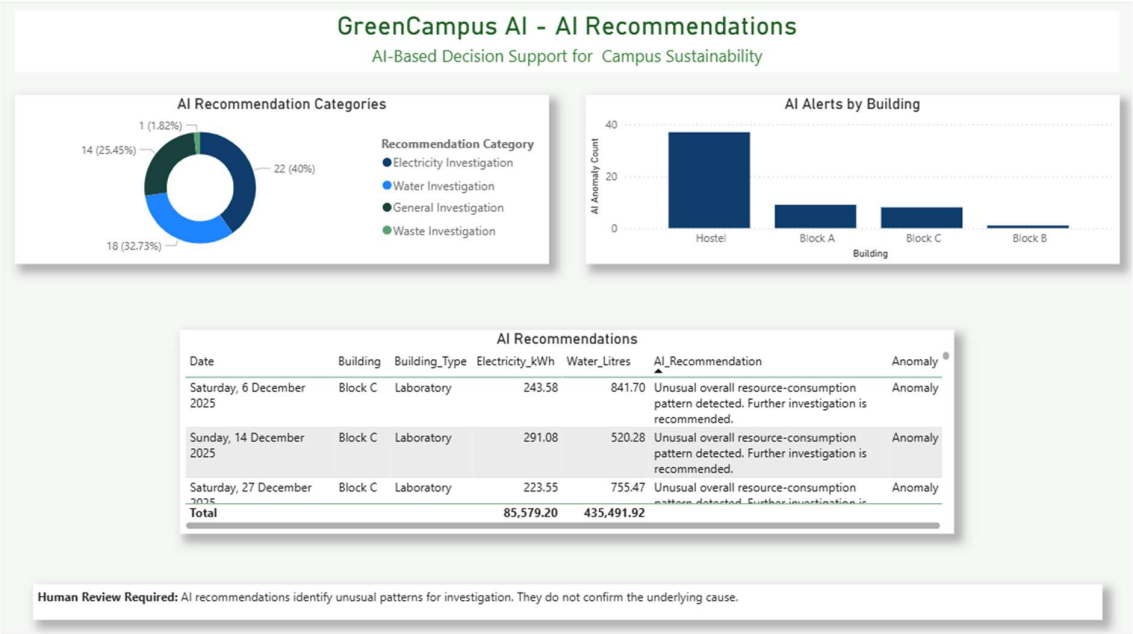


Figure 20.1 GreenCampus AI – AI Recommendations dashboard

21. AI Workflow

The complete project workflow can be represented as follows:

Simulated Campus Data

↓

Data Cleaning

↓

Exploratory Data Analysis

↓

AI Anomaly Detection

↓

Electricity Prediction

↓

AI Recommendation System

↓

Power BI Dashboard

↓

Human Review & Action

The workflow shows how raw data is converted into useful information and AI-supported recommendations.

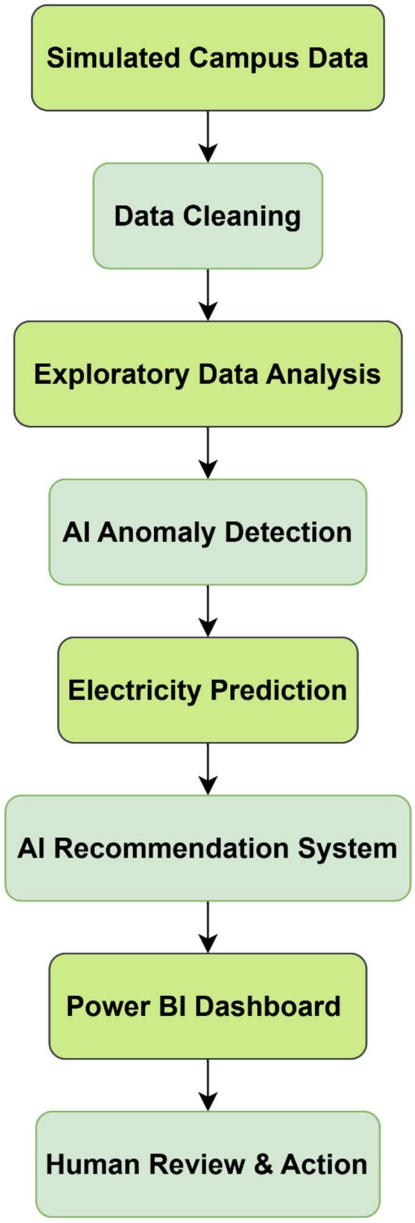


Figure 21.1 Overall GreenCampus AI system workflow

22. Role of AI in the Project

AI is not used only for visualization. It has specific roles in the system.

AI Function	Purpose
Pattern Detection	Understand resource-consumption patterns
Anomaly Detection	Identify unusual observations
Prediction	Estimate electricity consumption
Decision Support	Suggest areas for investigation

This is important because the internship guideline requires AI to have a clear and meaningful role in the solution.

23. Responsible AI

Responsible AI is an important part of GreenCampus AI.

The project considers **fairness, transparency, privacy, ethics and human oversight**, as required by the internship guideline.

23.1 Fairness

Different buildings can have different sizes, purposes and occupancy levels.

For example, a hostel may naturally consume more resources than a small classroom block.

Therefore, resource consumption should not be compared without considering the context of the building.

23.2 Transparency

The system does not simply provide an unexplained alert.

The dashboard shows:

- Building
- Date
- Resource consumption
- Anomaly status
- Recommendation

This helps users understand why an observation has been highlighted.

23.3 Privacy

The project uses simulated data.

No personal information about individual students, faculty members or staff is required.

23.4 Human Oversight

AI recommendations are used only for decision support.

An anomaly does not automatically prove the cause of a problem.

For example:

High water consumption → **investigate possible leakage or unusual usage**

rather than:

High water consumption → **water leak confirmed**

Final decisions should be made by appropriate campus staff after checking the actual situation.

24. Testing

The system was tested at different stages.

Dataset Testing

The dataset was checked for:

- Missing values
- Duplicate records
- Negative values
- Correct date range
- Valid building names
- Valid working-day and event-day values

AI Testing

The anomaly detection model was checked against unusual cases introduced into the simulated dataset.

Dashboard Testing

The Power BI dashboard was checked for:

- Correct KPI values
- Correct filters
- Correct building-level charts
- Correct anomaly counts
- Correct recommendation display

25. Results

The project produces the following outputs:

Data Analysis

The system provides:

- Building-wise resource analysis
- Time-based resource trends
- Occupancy-related analysis
- Temperature-related analysis
- Working-day and event-day analysis

AI Analysis

The system provides:

- Anomaly identification
- Anomaly scores
- Electricity predictions
- AI-generated investigation recommendations

Dashboard

Power BI provides an interactive interface for viewing:

- Resource consumption
- AI anomalies
- Building-level patterns
- Recommendation categories
- Individual anomaly records

26. Expected Impact

GreenCampus AI can support campus sustainability by providing a structured way to monitor and analyze resource consumption.

Expected benefits include:

- Better awareness of resource usage
- Easier identification of unusual consumption
- Data-supported investigation
- Better understanding of building-level patterns
- Support for sustainability planning
- Increased awareness among campus stakeholders

The project does **not claim a specific percentage reduction in electricity, water or waste**, because such an improvement would require testing the system using real campus data over time.

27. Limitations

The current prototype has some limitations.

1. Simulated Data

The project uses synthetic data rather than actual campus resource data.

2. Limited Resources

The current system focuses mainly on:

- Electricity
- Water
- Waste

3. No Real-Time Sensors

The prototype does not currently receive live readings from IoT sensors.

4. Anomaly Detection Does Not Identify Exact Causes

The AI identifies unusual patterns but cannot confirm whether the cause is a leakage, equipment problem, event or another factor.

5. Prediction Depends on Data Quality

Prediction performance can change when the model is trained using different or real-world datasets.

28. Future Scope

The project can be extended in the future.

1. Real-Time IoT Integration

Sensors could be connected to collect real-time:

- Electricity
- Water
- Waste
- Temperature data.

2. Real-Time Alerts

The system could send notifications when an unusual pattern is detected.

3. Mobile Application

A mobile application could allow administrators to monitor campus sustainability from anywhere.

4. More Prediction Models

Future versions could predict:

- Water consumption
- Waste generation
- Energy demand

5. Carbon Footprint Analysis

The system could calculate estimated carbon emissions from electricity and other activities.

6. More Advanced AI

Future versions could include conversational AI or retrieval-based systems to help users ask questions about campus sustainability data.

29. Conclusion

GreenCampus AI demonstrates how AI and data analytics can be applied to a real-world sustainability problem.

The project starts with simulated campus resource data and processes it through data cleaning and exploratory analysis. AI is then used for anomaly detection and electricity prediction. A recommendation component provides investigation suggestions for unusual resource-consumption patterns.

Power BI is used to present the results through an interactive two-page dashboard.

The project also considers responsible AI by addressing fairness, transparency, privacy and human oversight.

Overall, GreenCampus AI provides a conceptual and working prototype for using AI as a **decision-support tool for campus sustainability**. With real campus data, IoT integration and further testing, the system could be developed into a more practical sustainability monitoring platform.